

Cryptography

CRYPTOGRAPHIC ALGORITHMS - RSA Questions

- A. Callie wants to send the message $M = 13$ to Alice. Using Alice's public and private keys, calculate the ciphertext C , and the value for R when Alice recovers the message.
- B. Dexter wants to set up his own public and private keys. He chooses $p = 23$ and $q = 19$ with $e = 283$. Find d so that ed has a remainder of 1 when divided by $(p - 1)(q - 1)$.
- C. The attacker notices that Alice's public key is parametrised by $(n, e) = (391, 43)$. He quickly notices that p and q are such that $n = p * q$ are 17 and 23, respectively. How would the attacker find the value of Alice's decryption key?
- D. Why do we never let e be equal to 2?
- E. Jack is sending Tommy a message with RSA. The public key is 3, while N is 55. What is the value of d that Tommy must use to decrypt the message?
- F. If $p = 7$, $q = 11$, and $e = 17$. What is the public key? What is the value of d ? What is the private key?
- If $A = 0$, $B = 1$, $C = 2$, $D = 3$, etc What would the ciphertext of "HELLO" be?